Kvadratická nerovnice

$$ax^2 + bx + c < 0$$

$$ax^2 + bx + c > 0$$

$$ax^2 + bx + c \leq 0$$

$$ax^2 + bx + c \geq 0$$

Postup:

1. nerovnici upravíme na tvar, ve kterém je napravo 0

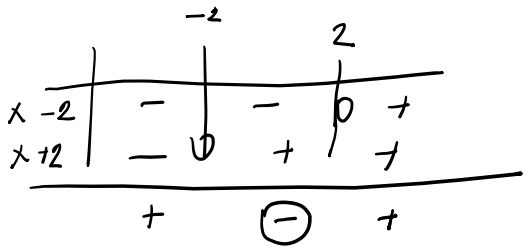
2. najdeme kořeny kvadratické rovnice a nerovnici upravíme na součinový tvar

3. pokud má rovnice záporný diskriminant, dosadíme do nerovnice jakékoliv číslo a posoudíme, zda nerovnost dává smysl.

Př. 1

$x^2 - 4 < 0$

$(x-2)(x+2) < 0$



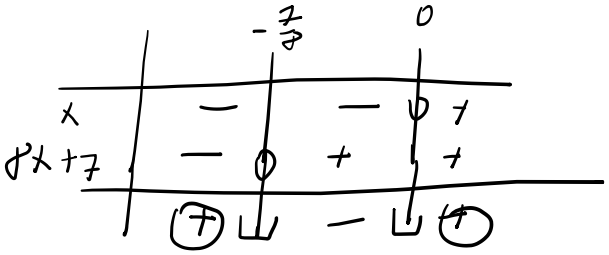
Př. 2

$(4x - 2)(2x + 3) \geq x - 6$

$8x^2 + 12x - 4x - 6 \geq x - 6$

$8x^2 + 7x \geq 0$

$x(8x + 7) \geq 0$



Př. 3

$3x^2 + 5x - 2 < 0$

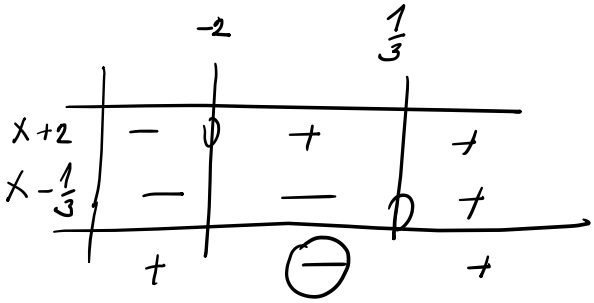
$D = 25 + 4 \cdot 3 \cdot 2 = 49$

$x_{1,2} = \frac{-5 \pm 7}{6} = \begin{cases} x_1 = -2 \\ x_2 = \frac{1}{3} \end{cases}$

$a(x - x_1)(x - x_2)$

$3(x+2)(x - \frac{1}{3}) < 0 \quad / : 3$

$(x+2)(x - \frac{1}{3}) < 0$



$K = (-2; \frac{1}{3})$

Př. 4

$x^2 - 15x + 56 \leq 0$

$D = 15^2 - 4 \cdot 56 = 1$

$x_{1,2} = \frac{15 \pm 1}{2} = \begin{cases} 8 \\ 7 \end{cases}$

$(x-8)(x-7) \leq 0$

$K = \langle 7, 8 \rangle$

Př. 5

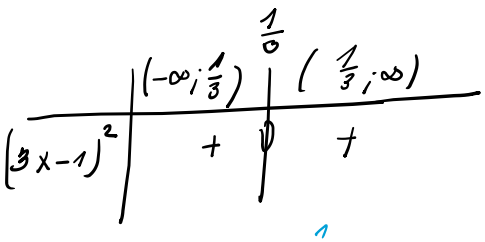
$9x^2 - 6x + 1 \leq 0$

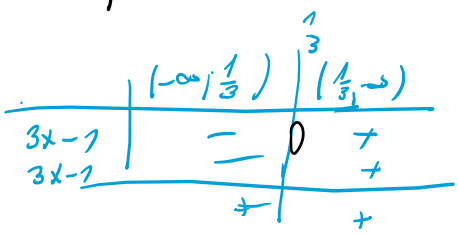
$D = 0$

$(3x-1)^2 \leq 0$

$x_{01} = \frac{1}{3}$

$K = \{\frac{1}{3}\}$





$9x^2 - 6x + 1 < 0$ $\quad K = \emptyset$

$9x^2 - 6x + 1 \geq 0$ $\quad K = \mathbb{R}$

$9x^2 - 6x + 1 > 0$ $\quad K = \mathbb{R} - \{\frac{1}{3}\}$

Př. 6

$x^2 + 5x + 6 > -1$

$x^2 + 5x + 7 > 0$

$D = -3$

$D < 0$ 1) nerovnice nemá řešení $K = \emptyset$

2) $K = \mathbb{R}$

$x = 0$ $\quad 0^2 + 5 \cdot 0 + 7 > 0$

$7 > 0$ pravda $\quad K = \mathbb{R}$

Př. 7

$-x^2 + 6x - 5 \geq 0$ $\quad / \cdot (-1)$

$x^2 - 6x + 5 \leq 0$

$(x-1)(x-5) \leq 0$

$K = \langle 1, 5 \rangle$